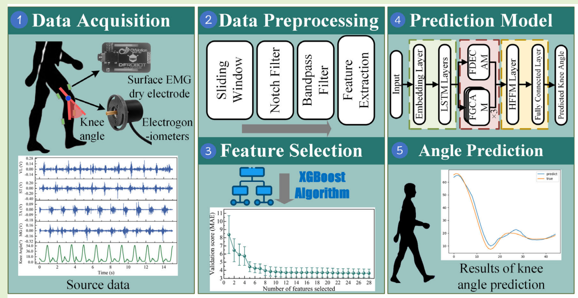
sEMG-Based Knee Angle Prediction: An Efficient Framework With XGBoost Feature Selection and Multiattention LSTM

Liuyi Ling, Liyu Wei, Bin Feng, Zhipeng Yu, and Long Wang



***Abstract*—Accurate prediction of lower limb joint angles is essential for enabling natural human–exoskeleton interaction in rehabilitation robotics. This study proposes a novel framework for knee joint angle prediction using surface electromyography (sEMG) signals, integrating an XGBoost-driven feature selection algorithm and a multiattention hybrid-enhanced long short-term memory (LSTM) network. First, sEMG signals were acquired from healthy participants during dynamic lower limb movements. After preprocessing, temporal and spectral features were extracted, after which the eXtreme Gradient Boosting (XGBoost) algorithm was applied to eliminate redundant features, reducing input dimensionality while maintaining predictive accuracy. Finally, the reduced features were fed into the proposed model, which leverages hybrid attention mechanisms to enhance temporal dependencies and feature relevance. The experimental results validate that the XGBoost-driven feature selection framework significantly minimizes redundancy in sEMG feature extraction. When evaluating the performance of joint angle prediction, the mean absolute error (MAE), root mean square error (RMSE), adjusted R^2, and Pearson correlation coefficient (CC) of the proposed model were 2.47°, 3.55°, 0.95, and 0.98, outperforming traditional machine learning (ML) algorithms and the benchmarks CNN, LSTM, TCN, and CNN-BiLSTM. The framework's superior computational efficiency and prediction accuracy highlight its potential for real-time implementation in exoskeleton systems, addressing critical limitations in existing control paradigms. This advancement paves the way for adaptive human–robot collaboration in clinical rehabilitation settings.**

***Index Terms*— Angle prediction, attention mechanism, deep learning (DL), feature selection, surface electromyography (sEMG).**

I. Introduction

During the process of gait rehabilitation assistance, lower limb exoskeleton rehabilitation robots (LLERos) require sophisticated intention recognition capabilities and adaptive control strategies to enable active patient participation and dynamic gait adaptation [1]. However, relying solely on predefined motion patterns to trigger robot movement can compromise stability and coordination. Therefore, a control system capable of modulating robot parameters, such as joint angles, in real time based on perceived movement intentions is crucial for achieving precise human–robot interaction and enhancing user comfort [2]. Common control signal sources for LLERos include physical sensing and bioelectrical signals. Physical sensors, such as inertial measurement units (IMUs) [3], pressure sensors [4], and visual feedback systems [5], offer robust antijamming capabilities and a high signal-to-noise ratio, making them effective for predicting human motion intentions. However, these signals often lag behind actual human movement, potentially reducing the accuracy of intention prediction. Surface electromyography (sEMG) signals, on the other hand, directly reflect muscle activity and provide a more immediate representation of human movement information [6]. With mature, noninvasive acquisition techniques, sEMG signals can precede actual limb

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The authors are with the School of Artificial Intelligence, Anhui University of Science and Technology, Huainan 232001, China (e-mail: 2022201848@aust.edu.cn).

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movement by 30–150 ms, enabling sEMG-based control to predict movement intention in advance [7], [8]. This predictive capability enhances the real-time performance and safety of human–computer interaction systems, making sEMG a widely adopted approach in recent research on human motion intention prediction.

In recent years, the prediction of human motion intentions based on sEMG has predominantly utilized traditional biological analysis, machine learning (ML) models, and deep learning (DL) models. For instance, Han et al. [9] integrated forward dynamics into a Hill-based muscle model to estimate human joint motion directly from sEMG, employing the Kalman algorithm for refinement. Yao et al. [10] developed EMG-driven control system for an ankle exoskeleton by incorporating a human muscle model. However, physiological models tend to be complex and have limited prediction effects, posing challenges for practical implementation. In contrast, ML approaches facilitate the construction of joint angle prediction models with greater ease. Typical models include linear regression (LR) [11], support vector machine regression (SVR) [12], random forests (RFs) [13], and artificial neural networks. Hahne et al. [11] compared the effects of LR and nonlinear regression techniques on synchronous and proportional myoelectric control of wrist movements. These ML methods rely on sEMG feature extraction to establish a correlation with joint angles, yet their accuracy heavily relies on signal quality, often resulting in significant robustness issues and prediction errors. Conversely, DL models have demonstrated substantial potential in sEMG-based joint angle prediction tasks. Zhang et al. [14] introduced a feature-guided convolutional neural network (CNN) to estimate knee joint motion using single-channel sEMG, which achieved higher estimation accuracy compared to the original CNN. The long short-term memory (LSTM) network has emerged as a powerful tool for sEMG analysis because of its ability to capture long-term dependencies in sequence data. Kim et al. [15] utilized LSTM as the central processor to estimate human joint positions and torques in real time from sEMG signals. Li et al. [16] used LSTM to extract sEMG features to establish the relationship between human lower limb flexion/extension and joint angles, achieving superior predictive performance over deep neural networks. Sun et al. [28] proposed a feature-based CNN-BiLSTM model to fuse sEMG and IMU data, resulting in reduced estimation errors compared to single-sensor approaches. Furthermore, the incorporation of attention mechanisms enables the extraction of key features from complex information while suppressing irrelevant details. Wang et al. [17] proposed an LSTM model with an attention mechanism for learning global sEMG features, significantly enhancing the accuracy and efficiency of online estimation. Wang et al. [18] also proposed a dSTA-EMGNet model based on spatial–temporal attention to enhance the prediction performance of the model for knee angles.

In the field of continuous prediction of human lower limb joint angles, existing methods still face several challenges. To enhance the accuracy of joint angle prediction, some researchers utilize a wide variety of sEMG features as the input feature vectors for neural networks, yet overlook the redundancy that may exist among these features. Therefore, how to effectively extract key features still needs further research. Gehlot et al. [20] proposed an explainable AI-driven feature selection framework combining filter, wrapper, and embedded methods to retain 50% of optimal features. Additionally, some studies focus solely on estimating current joint angle values, but neglecting the transmission response delay of the exoskeleton's mechanical structure [21], [22]. This study aims to continuously predict knee angle in advance and explore the impact of prediction time on the performance of joint angle prediction. Meanwhile, this study also aims to screen redundant sEMG features to reduce the dimensionality of the input data. The main contributions of this study include the following: 1) a feature selection algorithm based on XGBoost is designed to quantify the contribution of different sEMG features for knee angle prediction and eliminate redundant features, thereby reducing the number of required features; 2) we propose a joint angle prediction model, which combines the attention mechanism to more effectively model the channel interaction in sEMG and make full use of the historical time information; and 3) experiments on human subjects verify the reliability of our method, achieving an RMSE of 3.55°, and quantitatively analyzing the impact of different prediction times on prediction performance.

II. Materials and Methods

In our method, first, raw sEMG signals were acquired and subjected to necessary preprocessing steps and feature extraction. Subsequently, a feature selection algorithm based on XGBoost was applied to retain only the most informative features, thereby reducing dimensionality. Then, the selected sEMG feature data was divided into training set and test set. The proposed DL model was trained using the train data, and the best network parameters were obtained after cross-validation. Finally, the model was used for knee angle prediction.

A. DAQ Protocol

Six healthy participants without any gait impairments were recruited for the study. The participants had an average age of 24 ± 1.2 years and a standard body mass index of 23.44 ± 0.6 kg/m2. The Anhui University of Science and Technology Biomedical Research Ethics Committee approved the experiment (LW2023002). All participants provided informed consent prior to the commencement of the experiments and could be terminated at any time.

The study's experiments took place in a general living environment with normal temperature, humidity, and electromagnetic interference. The left leg was degreased using 95% medical alcohol, followed by a rinse with warm water. Before walking, the participants were equipped with wearable sensors to capture the activity of the left lower limb muscles and knee joint kinematic data. The sEMG signals were recorded using electromyographic electrodes (SEN0240; Zhiwei Robotics Corp., Shanghai, CN) from the four muscles in the left leg: the tibialis anterior (TA) muscle, the medial gastrocnemius (MG) muscle, the vastus lateralis (VL) muscle, and

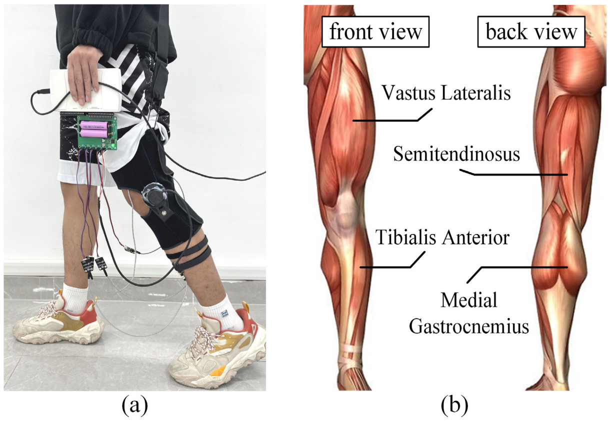

Fig. 1. Site for the acquisition of sEMG data from the left lower limb. (a) Actual sensor connection configuration. (b) Schematic representation of target muscle locations.

the semitendinosus (ST). The selection of these muscles was based on human lower limb muscle anatomy and the existing literature [29], as they play critical roles in lower limb function and movement. Fig. 1 shows the sEMG signal acquisition site of the left leg. Kinematic signals of the knee joint (sagittal plane only) were recorded using electrogoniometers (FS-360051C; Fusheng Electronic Technology Co., Ltd., Shanghai, CN) placed on the knee. The National Instruments (NI) USB-6210 data acquisition (DAQ) device, in collaboration with the NI-DAQmx software, was employed to capture and record the data from all sensors, thereby enabling the visualization and documentation of the collected information.

Each participant was asked to walk in a straight line on level ground at their self-selected pace, with each individual completing five walking trials, each lasting approximately 1 min. To mitigate the effects of muscle fatigue, each walking trial was separated by approximately 5 min of rest. The sampling frequency of the sEMG signal was set to 1000 Hz. In addition to the self-selected pace trials, we also conducted treadmill-based experiments to capture data at controlled walking speeds. Based on the experimental paradigms outlined in [30] and [31], we asked all subjects to repeat the data collection five times at each of four different speeds (1.2, 1.5, 1.8, and 2.1 km/h), which are commonly used in gait analysis studies.

B. Data Preprocessing and Feature Extraction

In the field of prosthetic control, where a robust interpretation of the user's intention is paramount, maintaining a clean sEMG signal is a prerequisite [32], [33]. To achieve this, it is essential to preprocess the raw data. Initially, a notch filter was applied to eliminate the 50 Hz power line interference and reduce motion artifacts. Subsequently, given that the effective frequency range of sEMG signals typically spans from 10 to 450 Hz, a bandpass filter (forth-order Butterworth) with cut-off frequencies at 10 and 450 Hz was employed to remove the majority of the low- and high-frequency noise [34], [35]. Fig. 2 presents a comparison of the sEMG signals before and after denoising. Goniometer signals were low-pass filtered (fourth-order Butterworth) at 10 Hz, respectively.

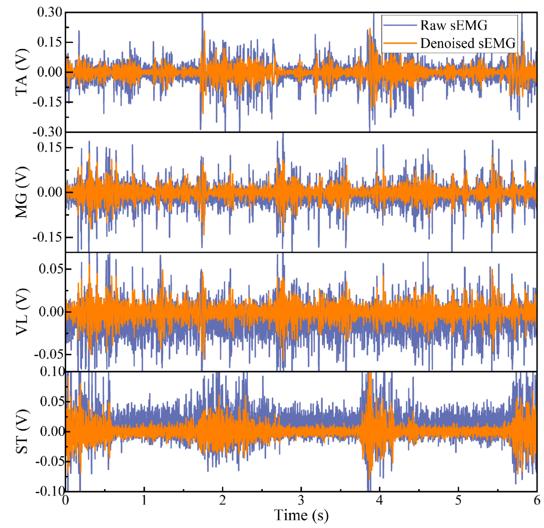

Fig. 2. Comparison of sEMG signals before and after denoising.

Building on methodologies established in prior work [19], [23], we selected and extracted a total of seven features: five from the time domain and two from the frequency domain. Time domain features encompassed mean absolute value (MAV), root mean square (rms), wavelength (WL), variance (VAR), and zero crossing (ZC). In the frequency domain, the features were the mean frequency (MNF) and median frequency (MDF). They are either energy or frequency related and have been widely used in medical and engineering practice. These seven sEMG features can be calculated as follows:

$$\mathrm{MAV} = \frac{1}{N}\sum_{i=1}^{N}|s_i| \tag{1}$$

$$\mathrm{rms} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}s_i^2} \tag{2}$$

$$\mathrm{WL} = \sum_{i=1}^{N-1}|s_{i+1} - s_i| \tag{3}$$

$$\mathrm{ZC} = \sum_{i=1}^{N-1}\left[\mathrm{sgn}\,(s_i \times s_{i+1}) \cap |s_i - s_{i+1}| \geq \mathrm{threshold}\right]$$

$$\mathrm{sgn}\,(s) = \begin{cases} 1, & \text{if } s \geq \text{threshold} \\ 0, & \text{otherwise} \end{cases} \tag{4}$$

$$\mathrm{VAR} = \frac{1}{N-1}\sum_{i=1}^{N}s_i^2 \tag{5}$$

$$\mathrm{MNF} = \sum_{i=1}^{M}f_i P_i \Bigg/ \sum_{i=1}^{M}P_i \tag{6}$$

$$\mathrm{MDF} = \frac{1}{2}\sum_{i=1}^{M}P_i \tag{7}$$

where the magnitude of the sEMG signal is represented by s, while N indicates the length of the sEMG signal. The frequency at the ith frequency bin is denoted by f_i, and the sEMG power spectrum at the ith bin is given by P_i. Additionally, M signifies the width of the frequency bin.

Given that sEMG data is sequential in nature, the sliding window technique was utilized to capture both temporal and spectral features, ensuring their continuity. To enhance real-time performance and reduce latency while extracting additional feature data, the time windows were implemented with an overlap. Consequently, a time window duration of 100 ms and a stride of 20 ms were established for extracting the aforementioned seven features. The labels for the knee joint angles were calculated from averaging the angles recorded within each window [36], [37]. The selected window size falls within the recommended optimal range for myoelectric signal control, as demonstrated in prior studies [38], [39] and is commonly used in the literatures [40], [41]. Consequently, the windows were refreshed at intervals of 20 ms, resulting in an angle estimation rate of 50 Hz.

Z-score normalization was implemented on each feature channel of each sEMG channel to maintain all feature values at the same amplitude and accelerate model convergence. The function for this transformation is depicted in the following equation:

$$Z = \frac{s - \mu}{\sigma} \tag{8}$$

where Z is the standardized value, s denotes the actual sample value, μ signifies the average of the data, and σ stands for the data's standard deviation.

C. XGBoost-Based Feature Selection Method

In this study, we employed an XGBoost-based feature selection method to identify the most informative features for our prediction task. XGBoost is an ML algorithm based on gradient boosting, renowned for its superior computational efficiency and robustness against overfitting [42]. The fundamental operation of XGBoost is based on the sequential addition and training of new trees, aimed at reducing the errors that remain from previous iterations.

XGBoost quantifies feature importance by evaluating the information gain contributed by each feature. The gain metric represents the average improvement in model accuracy across all decision tree splits where the feature is utilized. Features associated with higher gain values are thus identified as more influential predictors.

The sEMG feature selection algorithm operates as follows. After feature extraction, the initial feature set F is derived. First, subject-specific feature importance scores are computed using the training data for each participant. These scores are subsequently normalized to a range of [0, 1]. Next, each feature in F is ranked in descending order of importance based on the average value of feature importance across all subjects. The F_i in F with the lowest importance is iteratively removed to create the subfeature F_i. Consequently, the number of features in F decreases incrementally, as each feature undergoes screening. Finally, XGBoost regression models are trained on each subset, and the configuration achieving optimal predictive performance on the test set is retained as the final feature set.

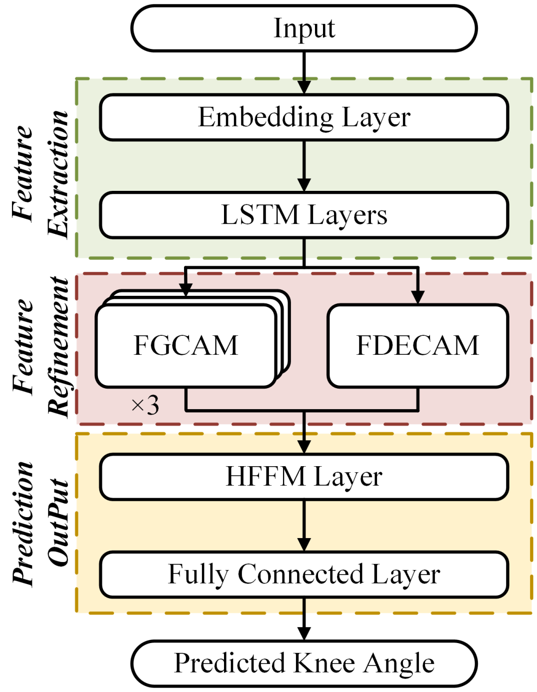

Fig. 3. Detailed illustration of the proposed channel attention-enhanced LSTM (CAELSTM) model for continuous knee angle prediction. The sEMG feature matrix is continuously input by passing through a sliding window, and the sEMG features are mapped to joint angles through three stages.

D. CAELSTM Model Structure

This section details the architecture of the proposed knee angle prediction model, CAELSTM. As illustrated in Fig. 3, the CAELSTM framework comprises three sequential stages: feature extraction, feature refinement, and prediction output.

1) Feature Extraction Stage: Following the aforementioned feature selection algorithm, the sEMG feature vectors extracted from 20 continuous sliding windows are combined to create the sEMG feature time-series matrix S, which serves as the input to the DL model. This input can be expressed by the following equation:

$$S = \begin{bmatrix} s_1^1 & s_2^1 & \cdots & s_W^1 \\ s_1^2 & s_2^2 & \cdots & s_W^2 \\ \vdots & \vdots & \vdots & \vdots \\ s_1^M & s_2^M & \cdots & s_W^M \end{bmatrix} \tag{9}$$

where the set $s_W = (s_W^1, s_W^2, \ldots, s_W^M)$ is the M feature value at the Wth window.

1) *CNN Embedding Layer:* The embedding layer employs Conv1D to convert the sequence of sEMG features from the C channel to the higher dimensions of C', thus enabling the network to learn more complex and abstract patterns.
2) *LSTM Layer:* LSTM network is an advanced type of recurrent neural network (RNN) that incorporates memory components, building upon the foundational structure of RNN [43]. This makes LSTM particularly well-suited for processing time series data, here we apply LSTM to sEMG feature sequences.

The LSTM architecture identifies the cell state utilizing three control gates: the forget gate (f_t), the input gate (i_t), and the output gate (o_t). The forget gate is responsible for deciding the extent to which the previous cell state (c_{t-1}) is preserved in the current cell state (c_t). It outputs a value between 0 and 1, which indicates that $c_t - 1$ should be completely forgotten, and 1 indicates that it should be fully retained. The input gate controls how much of new candidate cell state ($\tilde{c}_t$) should be added to the c_t. The output gate controls how much of c_t should be output as the hidden state (h_t). The calculation methods of the three gates are as follows:

$$f_t = \sigma\left(W_f \cdot \left[h_{t-1}, x_t\right] + b_f\right) \tag{10}$$
$$i_t = \sigma\left(W_i \cdot \left[h_{t-1}, x_t\right] + b_i\right) \tag{11}$$
$$o_t = \sigma\left(W_o \cdot \left[h_{t-1}, x_t\right] + b_o\right). \tag{12}$$

Among then, σ denotes the sigmoid activation function, which outputs values within the interval (0, 1); $[h_t - 1, x_t]$ is a vector composed of $h_t - 1$ and x_t; and the weight matrices W_f, W_i, and W_o correspond to the forget gate, input gate, and output gate, respectively. Similarly, the bias vectors b_f, b_i, and b_o are associated with the forget gate, input gate, and output gate, respectively.

2) Feature Refinement Stage: Recent studies have suggested incorporating an attention mechanism into channel selection. This approach aims to enable the network to autonomously discern the most crucial information, thereby enhancing the performance of key features. Therefore, this article presents a channel attention fusion network composed of fine-grained channel attention and frequency-domain enhanced channel attention. The structural design is illustrated in Fig. 4.

a) Fine-grained channel attention module: In recent years, squeeze-and-excitation network (SE-Net) is cited to better model channel features in multivariate time series [44]. The SE-Net employs a fully connected layer (FC) to gather global information but falls short in engaging with local data, leading to imprecise weighting of various channel features. To address this, we propose a fine-grained channel attention mechanism designed to seamlessly combine both global and local information. This approach ensures a more rational distribution of weights, thereby extracting more valuable features. The fine-grained channel attention module (FGCAM) is mainly applied in the feature refinement stage to perform further feature fine-tuning on the feature maps from the LSTM layer.

Given a feature map $F \in \mathbb{R}^{C\times L}$ as input, FGCAM infers a 1-D channel attention map $w \in \mathbb{R}^{C\times 1}$ as illustrated in Fig. 4, where L and C are the length of time and the number of channels. Unlike SE-Net, which directly performs squeeze operation on the input feature map, the two branches depthwise and pointwise first extract intrachannel information and interchannel information on the input feature map, and then sum and fuse them. In order to compute channelwise attention efficiently and finely, we compress the temporal information of each channel by global max pooling and global average pooling to generate two channel descriptors $s_{\max} \in \mathbb{R}^{C\times 1}$ and $s_{\text{avg}} \in \mathbb{R}^{C\times 1}$. After that, we adopt an adaptive fusion strategy to aggregate two different spatial context descriptors. The achievement of dynamic fusion is facilitated by a trainable coefficient, as depicted in the following illustration:

$$z = \sigma(\varepsilon) \times \sigma(s_{\max}) + (1 - \sigma(\varepsilon)) \times \sigma\left(s_{\text{avg}}\right) \tag{13}$$

where σ denotes the sigmoid function and z denotes the final channel descriptor. This approach leads to channel descriptors with richer information while reducing redundancy. Then, to avoid increasing model complexity while capturing local cross-channel interaction information, we employ a fast 1-D convolution operation with a kernel size of k, wherein the kernel size k denotes the scope of the local cross-channel interaction, i.e.,

$$z_l = \text{Conv 1D}_k(z) \tag{14}$$

Here, z_l signifies the local data, and k stands for the size of the kernel. To acquire global channel data and enhance the capacity for global information representation, we employ a Conv2D operation where both the input and output channels are set to C, and the convolution kernel dimensions are 1×1 for implementation to achieve, i.e.

$$z_g = \text{Conv 2D}_{1\times 1}(z) \tag{15}$$

where z_g denotes the global information. To facilitate effective interaction between global and local information, we capture the correlations at multiple granularities through cross-correlation operations, i.e.

$$M = z_l \cdot z_g^{\mathrm{T}} \tag{16}$$

where M denotes the correlation matrix. Subsequently, we derive row and column data from this matrix and sum the elements to get the final weight vector, as illustrated below

$$w = \sigma\left(\sum_j^c M_{i,j} + \sum_j^c M^{\mathrm{T}}{}_{i,j}\right), \quad i \in 1, 2, 3, \ldots, c \tag{17}$$

where c is the number of channels. Ultimately, this methodology selectively highlights relevant features, downplays less significant ones, and balances a more accurate distribution of weights for features pertinent to angle prediction. Finally, the obtained weights are multiplied with the input feature map as follows:

$$F_{\text{att1}} = w \otimes F \tag{18}$$

where F is the input feature map, $\otimes$ denotes elementwise multiplication, and F_{att1} is the final output feature map.

b) Frequency-domain enhanced channel attention module : Some studies have indicated that improving the frequency modeling performance of time-series prediction models can result in enhanced outcomes [45], [46]. In this case, we incorporate the frequency-enhanced channel attention mechanism, which leverages the discrete cosine transform (DCT) to capture interchannel frequency dependencies. By employing DCT, we transform temporal signals into the frequency domain, thereby extracting spectral characteristics and mitigating high-frequency noise.

First, the sEMG feature sequence $F \in \mathbb{R}^{c\times L}$ is split along the channel dimension into c discrete single-channel variables $\{s_0, s_1, \ldots, s_{c-1}\}$, where $S_i \in \mathbb{R}^{1\times L}$, $i \in \{0, 1, \ldots, c-1\}$.

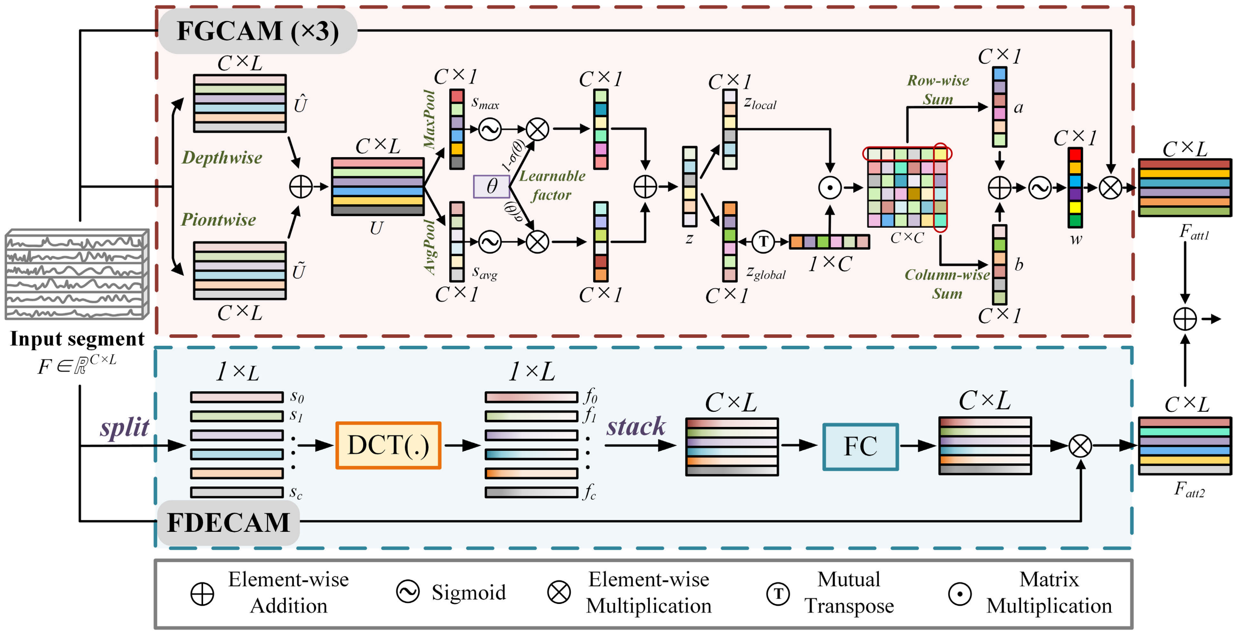

Fig. 4. Detailed internal structure of the feature refinement stage.

Then, we perform 1-D DCT along the temporal dimension to convert each channel of F to the frequency domain. Typically, the basic function of 1-D DCT is

$$f^l = \sum_{i=0}^{L_s-1} x_i \cos\left(\frac{\pi l}{L_s}\left(i+\frac{1}{2}\right)\right) \tag{19}$$

where $x \in \mathbb{R}^L$ is the input single-channel sEMG feature variable, $L \geqslant 1$ is the length of x, $l \in \{0, 1, \ldots, L_s - 1\}$, and $f^l \in \mathbb{R}^L$ is the 1-D DCT frequency spectrum.

Next, we stack each frequency channel vector along the channel dimension to get tensor $F_{\text{freq}} \in \mathbb{R}^{C\times L}$ which represents the spectrum of F. The fully connected network structure is then used to obtain the frequency channel attention, denoted as $F_{\text{att2}} \in \mathbb{R}^{C\times L}$. This is used to establish the frequency dependence between the different channels and enhance the model's capacity for nonlinear mapping and local pattern learning.

c) Feature fusion procedure: The output is a new feature map that contains the combined information from both attention modules. This fused feature map is expected to carry richer and more nuanced representations of the input data.

3) Prediction Output Stage: This critical stage is designed to synthesize the enriched features from the previous stages, leveraging the temporal dynamics captured by the LSTM network, enhanced by the channel attention mechanism, to produce accurate and reliable estimates of the knee joint angles.

a) Historical Feature Fusion Module: In order to obtain the importance of the hidden information of the historical moments of the sEMG signal for the joint angles, this article introduces the temporal pattern attention (TPA) network [47]. The TPA dynamically captures the impact of various factors on joint angle predictions by utilizing a series of 1-D convolutional kernels to extract time-invariant temporal patterns from the hidden state matrix produced in previous stages. A scoring function then assesses the correlation between the current hidden state vector and the historical time information. This process enhances the sensitivity of the input at pivotal historical moments, improving the model's ability to retain salient temporal features. Ultimately, a more robust feature vector is generated, which incorporates historical information. The structure of the TPA network is depicted in Fig. 5.

The input sEMG feature sequence is first subjected to two stages: feature extraction and feature refinement. The objective of these stages is to obtain the hidden state h_i at each time step, thereby generating the hidden state matrix $H = \{h_{t-w}, h_{t-w+1}, \ldots, h_t\}$, where the row vector represents a state factor at all time steps and the column vector represents all state factors at a certain time step. The convolutional kernels are applied to the row vectors of the hidden layer's output matrix to generate the temporal feature matrix, denoted as H^C, where $H^C \in \mathbb{R}^{m\times k}$. This is calculated as follows:

$$H^C_{i,j} = \sum_{l=1}^{w} H_{i,(t-w-1+l)} \times C_{j,T-w+l}. \tag{20}$$

We use the following evaluation function f to calculate the relevance:

$$f\left(H_i^C, h_t\right) = \left(H_i^C\right)^{\mathrm{T}} W_a h_t \tag{21}$$

where H_i^C is the ith row of H^C, and $W_a \in \mathbb{R}^{k\times m}$. The attention weight α_i is obtained as follows:

$$\alpha_i = \text{sigmoid}\left(f\left(H_i^C, h_t\right)\right). \tag{22}$$

Then, the row vectors of H^C are weighted by α_i to obtain the vector $v_t \in \mathbb{R}^k$

$$v_t = \sum_{i=1}^{m} \alpha_i H_i^C. \tag{23}$$

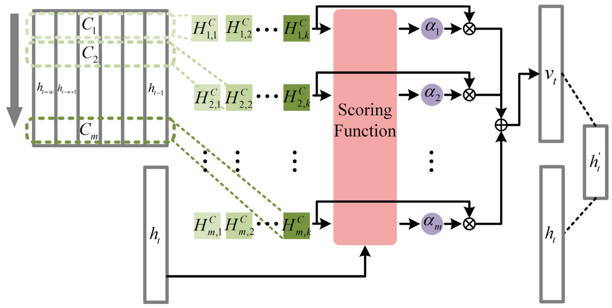

Fig. 5. Detailed internal structure of the HFFM module.

Finally, we integrate v_t and h_t to yield the final weight vector h_t'

$$h_t' = W_h h_t + W_v v_t \tag{24}$$

where $h_t, h_t' \in \mathbb{R}^m$, $W_h \in \mathbb{R}^{m\times m}$, and $W_v \in \mathbb{R}^{m\times k}$.

b) Fully Connected Layer : The feature vector derived from the historical feature fusion module (HFFM) is fed into the final FC layer, which utilizes the ReLU activation function. This layer outputs the estimated knee angle as its prediction.

E. Performance Evaluation

To develop a dependable angle prediction model, each participant's data was partitioned into training and testing subsets with a ratio of 7:3. The feature selection process was conducted on the training subset, which was further split into a training subset and a validation subset with a ratio of 4:1. Following this, k-fold cross-validation was utilized to train the XGBoost model and determine the final ranking of feature importance. The model's performance was then evaluated using the test subset to obtain the final metrics.

To evaluate the predictive performance, we employed several metrics, including mean absolute error (MAE), root square mean error (RMSE), adjusted R^2, and Pearson correlation coefficient (CC). MAE effectively reflects the size of the prediction error, while RMSE measures discrepancies between actual and predicted values. To mitigate the potential influence of sample size on the results, we use adjusted R^2 as a supplementary measure. The CC indicates the level of congruence between the actual and predicted shapes of the knee angle. The following equations illustrate the calculation formulas:

$$\text{MAE} = \frac{1}{m}\sum_{k=1}^{m}\left|\theta_k' - \theta_k\right| \tag{25}$$

$$\text{RMSE} = \sqrt{\frac{1}{m}\sum_{k=1}^{m}\left(\theta_k' - \theta_k\right)^2} \tag{26}$$

$$R^2 = 1 - \frac{\sum_{k=1}^{m}\left(\theta_k' - \theta_k\right)^2}{\sum_{k=1}^{m}\left(\theta_k - \bar{\theta}\right)} \tag{27}$$

$$\text{adjusted } R^2 = 1 - \frac{\left(1 - R^2\right)(m-1)}{m-p-1} \tag{28}$$

$$\text{CC} = \frac{\text{cov}\left(\theta' - \theta\right)}{\sigma'\sigma} \tag{29}$$

where θ denotes the actual joint angle, θ' signifies the predicted joint angle, $\bar{\theta}$ represents the mean of the actual joint angles, and m is the length of the angle sequence. The term $\text{cov}(\theta' - \theta)$ quantifies the covariance between the actual and predicted angles. The symbol σ' denotes the standard deviation of the predicted angles, σ stands for the standard deviation of the actual angles, and p denotes the count of features.

F. Experimental Details

Utilizing an NVIDIA GeForce RTX 3050 GPU, all model trainings were performed with a batch size of 64. We used a combination of Adam with betas of (0.9, 0.99) and a learning rate reduction on CosineAnnealingLR optimizes all used models. The initial learning rate for the model was set at 0.001, which was subsequently adjusted by the CosineAnnealingLR based on the training average loss. To prevent overfitting, early stopping was employed, whereby the model training was terminated prematurely if the accuracy did not improve over 7 epochs in the validation set. As the loss function, we adopted the mean squared error (MSE) to assess the difference between the predicted and actual angle values. The MSE is defined as follows:

$$\text{loss}\left(y, \hat{y}\right) = \left(y - \hat{y}\right)^2 \tag{30}$$

where loss is the loss function, y is the actual angle, and $\hat{y}$ is the predicted angle.

In order to demonstrate the effectiveness of the proposed model, a comparison is made between four traditional ML methods (i.e., LR [11], RF [13], [23], SVR [12], and XGBoost [24]) and four DL models (i.e., CNN [25], LSTM [26], TCN [27], and CNN-BiLSTM [28]) that have been previously employed in the field of sEMG-based joint angle prediction. All models are trained in the same way.

One-way analysis of variance (ANOVA) is used to perform the statistical analysis. The normality of the prediction outcomes is ascertained using the Shapiro–Wilk test. Subsequent to the one-way ANOVA, a post hoc analysis is performed using Tukey's multiple comparisons test to determine the statistical disparities among various prediction models. A significance level of $p = 0.05$ is uniformly applied across all statistical assessments.

III. Results

A. Results of Feature Selection Method

Feature selection plays a crucial role in constructing ML models by reducing input dimensionality while preserving predictive accuracy, thus minimizing computational resource consumption. Fig. 6 shows the ranking of feature importance. XGBoost was then used to train various feature subsets, with MAE as the evaluation metric to identify the optimal subset size. Fig. 7 depicts the outcomes of the XGBoost feature selection process. When the subset was reduced to 14 features, the MAE decreased to 4.51°. Further increases in the subset size did not significantly reduce MAE. Therefore, the subset with 14 features was selected as the final choice.

The specific selected features are shown in Table I. The data indicates that the muscles which produced the greatest

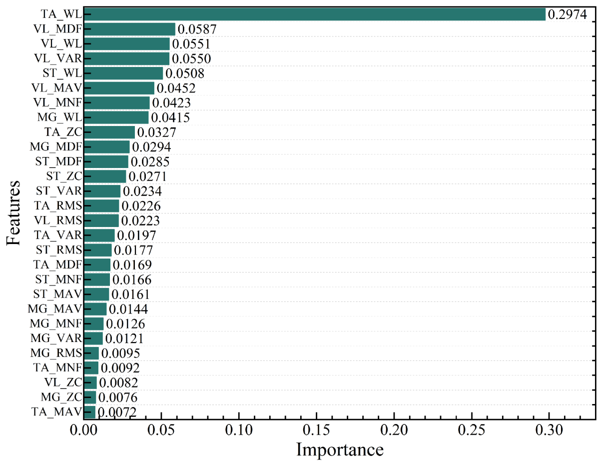

Fig. 6. Feature importance ranking in XGBoost.

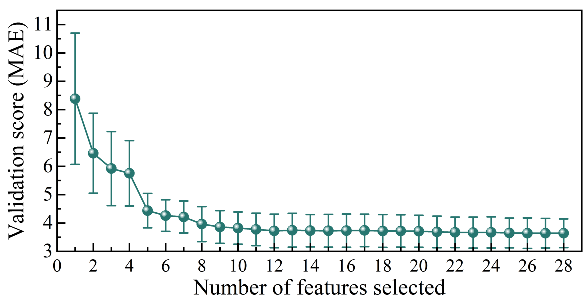

Fig. 7. Relationship between the number of feature variables and the MAE.

TABLE I
SPECIFIC SELECTED sEMG FEATURES

Associated lower limb muscles	sEMG features
Tibialis Anterior (TA)	WL, ZC, RMS
Medial Gastrocnemius (MG)	WL, MDF
Vastus Lateralis (VL)	WL, MDF, VAR, MNF, MAV
Semitendinosus (ST)	WL, MDF, ZC, VAR

number of sEMG signals are the VL and the ST muscles. The two main features identified are the WL and MDF features.

B. Impact of Feature Selection on Angle Prediction

To demonstrate the effectiveness of the proposed feature selection algorithm, sEMG feature sets before and after selection were used as the input for the LR, RF, SVR and XGBoost models to predict knee joint angle. A one-way ANOVA was conducted to evaluate the statistical significance of the results across the models. As shown in Fig. 8, no significant differences were observed in the predictive performance of the four models before and after feature selection ($p > 0.05$). This suggests that the feature selection algorithm effectively diminishes the dimensionality of the model's input features without sacrificing predictive accuracy.

C. Analysis of CAELSTM Model Results

Table II presents the performance evaluation of the proposed model for each subject on their self-selected walking speeds. As shown in Table II, for the experimental results of six healthy subjects, the average MAE, RMSE, adjusted R^2, and CC are 2.47°, 3.55°, 0.95, and 0.98, respectively. Among them, subject S03 achieved the lowest average adjusted R^2 and subject S01 achieved the highest value of 0.98. In general, the lowest adjusted R^2 is still higher than 0.91, and the estimated angle error is mainly concentrated in the limited range of (-3.8°, 3.8°), indicating that the model has good prediction performance. Table III presents the performance evaluation of the proposed model across different walking speeds. There is variability in model performance across different subjects, as evidenced by the differing RMSE and adjusted R^2 values. For example, subject S02 consistently exhibited higher RMSE values at all speeds.

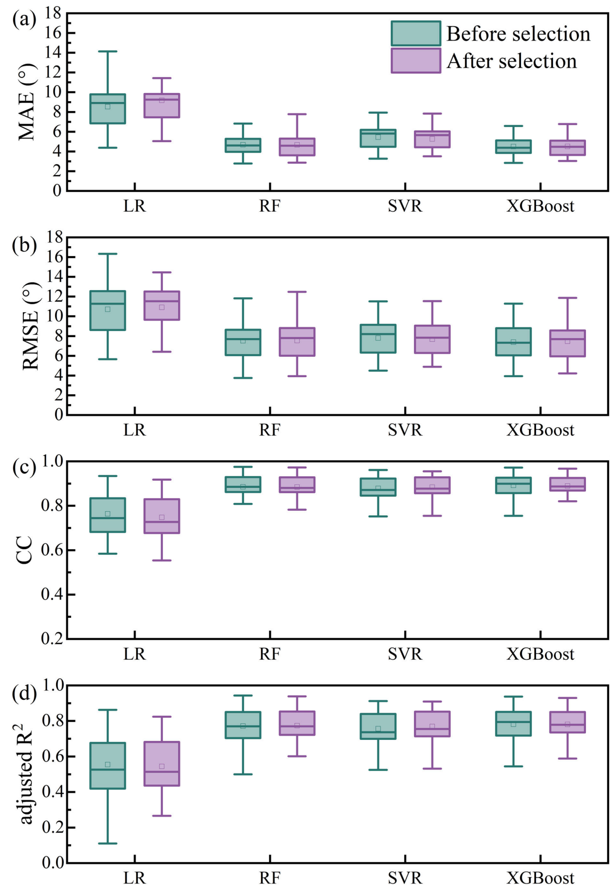

Fig. 8. Performance on different models before and after feature selection, respectively. (a) MAE. (b) RMSE. (c) Adjusted R^2. (d) CC.

TABLE II
KNEE ANGLE PREDICTION PERFORMANCE EVALUATION OF THE PROPOSED CAELSTM ON THE TEST SET OF ALL SUBJECTS

Subject	MAE (°)	RMSE (°)	adjusted R^2	CC
S01	1.98	2.95	0.98	0.99
S02	2.19	2.89	0.97	0.99
S03	2.62	3.94	0.92	0.96
S04	2.49	3.71	0.95	0.98
S05	2.72	3.88	0.94	0.98
S06	2.84	3.95	0.93	0.97
Average	2.47	3.55	0.95	0.98

To reveal the performance differences, we evaluate our approach with other methods. The results for each network

TABLE III
PERFORMANCE OF THE PROPOSED MODEL UNDER DIFFERENT SPEEDS OF DIFFERENT SUBJECTS

Subject	Speed 1.2 km/h		Speed 1.5 km/h		Speed 1.8 km/h		Speed 2.1 km/h	
	RMSE	adjusted R^2	RMSE	adjusted R^2	RMSE	adjusted R^2	RMSE	adjusted R^2
S01	3.71	0.95	2.78	0.97	3.41	0.96	3.74	0.96
S02	4.04	0.94	5.88	0.84	4.34	0.94	4.53	0.92
S03	4.28	0.90	4.99	0.91	4.47	0.93	4.05	0.90
S04	3.97	0.93	2.87	0.97	2.73	0.97	3.16	0.96
S05	4.14	0.92	3.70	0.94	2.95	0.97	3.40	0.94
S06	4.85	0.90	4.18	0.93	3.92	0.92	3.60	0.94
Average	4.17	0.92	4.07	0.93	3.64	0.95	3.75	0.94

TABLE IV
KNEE ANGLE PREDICTION RESULTS FOR DIFFERENT MODELS

Method	MAE (°)	RMSE (°)	adjusted R^2	CC
SVR	5.26	7.68	0.77	0.88
RF	4.67	7.55	0.77	0.88
XGBoost	4.51	7.47	0.78	0.89
CNN	4.00	5.38	0.88	0.95
LSTM	2.95	4.17	0.93	0.97
TCN	2.86	4.32	0.92	0.97
CNN-BiLSTM	2.81	4.37	0.91	0.96
Our model	2.47	3.55	0.95	0.98

are from experiments on the same test set, and the predictive performance is calculated by averaging the results over all subjects. Table IV presents the mean values of the MAE, RMSE, adjusted R^2, and CC of the CAELSTM model for all six healthy subjects. Our model achieves better predictions than other models.

To further evaluate model performance, we visualized the 6-s knee joint angle trajectories from the test sets of subject S05, comparing predictions from four DL architectures: CNN, LSTM, TCN, and CNN-BiLSTM. Traditional ML models were excluded due to their inferior performance in preliminary evaluations. As illustrated in Fig. 9, the proposed CAELSTM model exhibits significantly reduced fluctuations in predicted angles compared to other models, aligning closely with the ground-truth trajectory. This visual analysis highlights the enhanced stability and precision of CAELSTM, with minimal deviations observed across dynamic motion phases. Complementing this, Fig. 10 quantifies the comparative performance using RMSE and adjusted R^2 metrics. The proposed CAELSTM achieves the lowest RMSE (left panel), underscoring its superior prediction accuracy, while also securing competitive adjusted R^2 values (right panel), indicative of robust explanatory power.

Fig. 11 illustrates the prediction results of the proposed model under different walking speeds. It can be seen that the superior performance of our proposed model is obvious in all subjects, while CNN, LSTM, and CNN-BiLSTM prediction models will have large prediction deviations, and the effect of LSTM prediction model is second only to CAELSTM model.

D. Ablation Analysis of Model Components

To evaluate the effectiveness of the proposed CAELSTM model, an ablation study was conducted on the collected dataset. The goal of this study is to assess the impact of different model components on the prediction of joint angles 40 ms into the future. Specifically, the performance of five

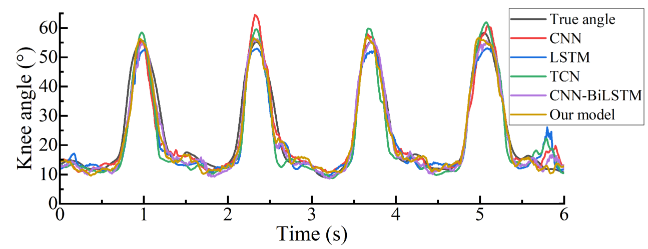

Fig. 9. Visual comparison of knee angle prediction results based on CNN, LSTM, TCN, CNN-BilSTM, and CAELSTM.

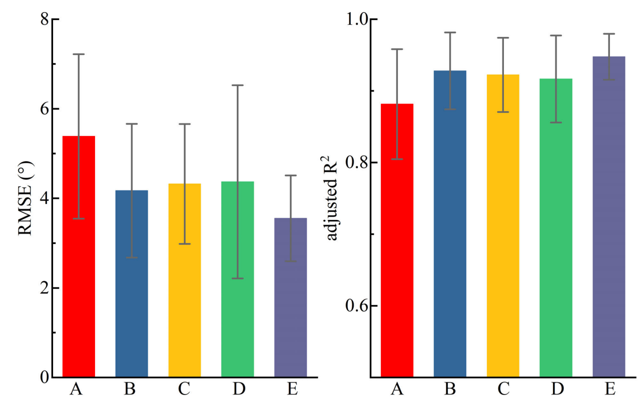

Fig. 10. Comparative evaluation of knee angle prediction models (A: CNN, B: LSTM, C: TCN, D: CNN-BiSTM, E: CAELSTM) using RMSE (left) and adjusted R^2 (right).

TABLE V
RESULTS OF THE ABLATION STUDY

Model	MAE (°)	RMSE (°)	adjusted R^2	CC
Base-NoAtt	2.96	4.49	0.89	0.95
Base-FGCAM	2.89	4.48	0.90	0.96
Base-FDECAM	2.86	4.42	0.90	0.96
Base-Refine	2.62	3.84	0.94	0.97
CAELSTM	2.47	3.55	0.95	0.98

model architectures was compared: the baseline model with only the feature extraction stage (Base-NoAtt), a baseline model with FGCAM modules integrated (Base-FGCAM), a baseline model with frequency-domain enhanced channel attention module (FDECAM) integrated (Base-FDECAM), and a CAELSTM model without HFFM (Base-Refine).

From the results in Table V, it can be seen that the proposed model has a significant improvement in RMSE and adjusted R^2 compared with Base-NoAtt model. This improvement underscores the efficacy of further refining the hidden features extracted in the initial feature extraction stage, coupled

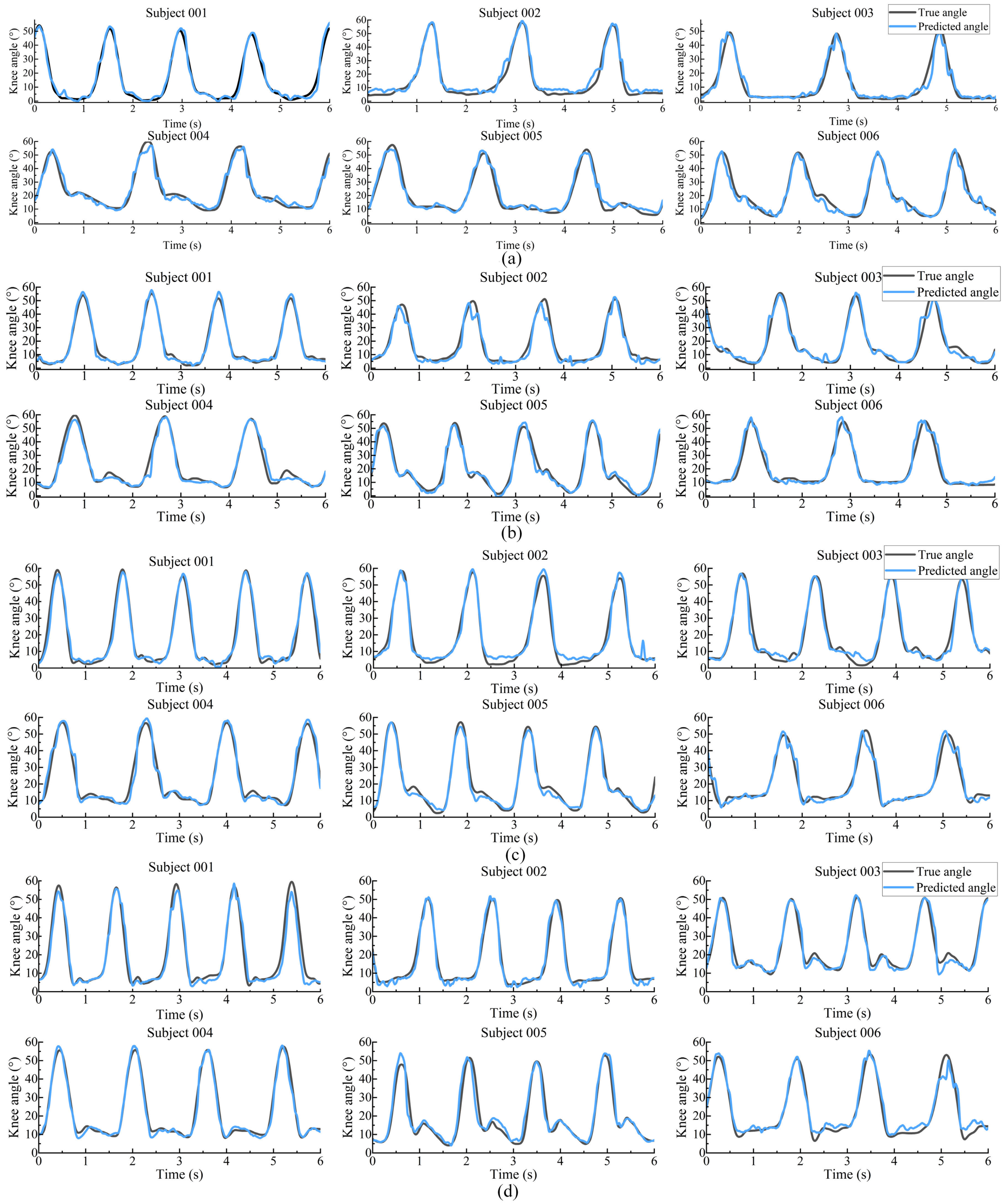

Fig. 11. Comparison of predicted results across different participants at varying walking speeds. (a) 1.2 km/h. (b) 1.5 km/h. (c) 1.8 km/h. (d) 2.1 km/h.

with the modeling of dependencies from historical temporal data, which together facilitate more precise predictions of joint angles. In addition, in the feature refinement stage, the study conducted comparisons by implementing single attention modules, namely FGCAM and FDECAM, individually. These models exhibited reductions of 0.07° and 0.1°, respectively, in the MAE, and enhancements of 0.008 and 0.01 in the adjusted R^2, as opposed to the Base-NoAtt model that did not incorporate an attention mechanism for feature refinement. These observations substantiate the significant role of attention mechanisms in augmenting model performance. Compared to the model without HFFM module, CAELSTM reduces MAE by 0.15° and RMSE by 0.29°.

E. Performance Evaluation for Different Prediction Times

To assess the impact of prediction horizon on model performance and to determine the boundaries of effective prediction, we selected five discrete time intervals: 40, 80, 120, 160, and 200 ms. Fig. 12 illustrates the average performance across these prediction times. It is evident that the predictive accuracy

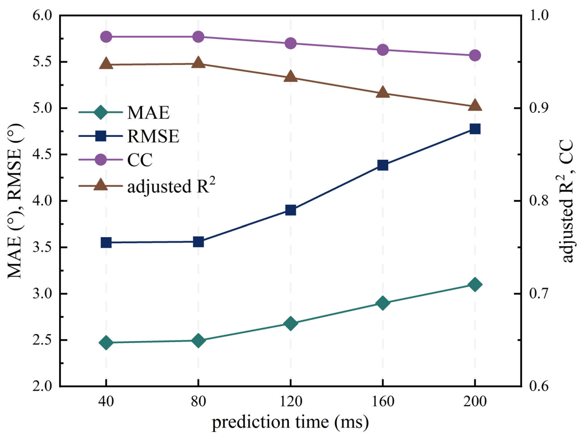

Fig. 12. Performance changes of the proposed model on different prediction durations.

declines with increasing prediction length, with the MAE exceeding 3° at the 200 ms mark, which is significantly poorer compared to the other four intervals. This threshold suggests a predictive boundary: to maintain model performance, the prediction duration should ideally be less than 200 ms.

Fig. 13 compares the performance of LSTM and CAELSTM models across the different prediction intervals, utilizing RMSE and adjusted R^2 as metrics. It can be observed that there is no significant difference between the LSTM and CAELSTM models across all prediction intervals ($p > 0.05$). Moreover, as the prediction time increases, the average RMSE for both models shows an overall upward trend. Specifically, at 40 ms, the RMSE for LSTM and CAELSTM is 4.17° and 3.55°, respectively; at 80 ms, it is 4.45° and 3.56°; at 120 ms, it is 4.75° and 3.90°; and at 160 ms, it is 4.86° and 4.39°. Conversely, the average adjusted R^2 demonstrates a downward trend, with values of 0.93 and 0.95 at 40 ms, 0.91 and 0.95 at 80 ms, 0.90 and 0.93 at 120 ms, and 0.90 and 0.92 at 160 ms for LSTM and CAELSTM, respectively. These results also indicate that the proposed CAELSTM model outperforms the LSTM model across all prediction durations.

F. Analysis of Channel Drop Study

In this study, we designed a channel drop study to evaluate the effect of different muscle combinations on the performance of joint angle prediction. We formed six different combinations of muscle channels by combining the muscles corresponding to the four channels in the thigh and calf regions: VLST, TAMG, VLTA, STMG, VLMG, and STTA. Among them, ALL represents that all four channels are used. Table VI presents the prediction performance at an average prediction delay of 40 ms. The selected sEMG features are determined based on the results in Table I.

Among the six channel combinations, the RMSE of VLST and STTA is 4.54° and 4.55°, respectively, which is 0.98° and 1.001° higher than that of ALL configuration, and the increase is relatively small. In contrast, the RMSE of VLTA and VLMG is 6.95° and 6.67°, respectively. Compared with the ALL configuration, the error increases by 3.40° and 3.12°, respectively, showing a larger error increase.

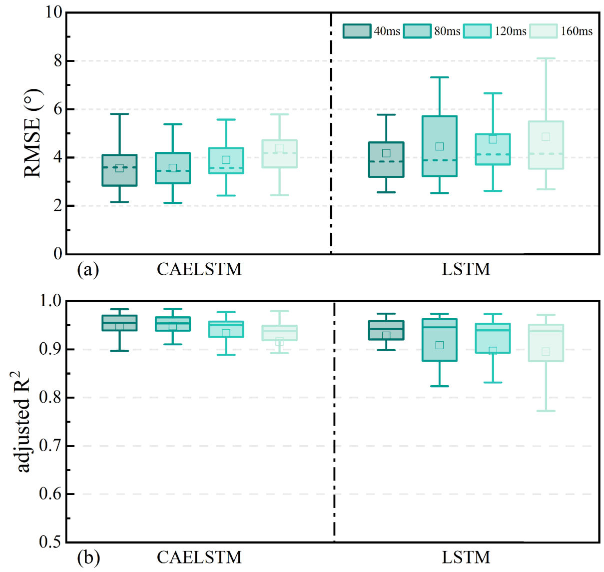

Fig. 13. Performance comparison of CAELSTM and LSTM on different prediction durations.

TABLE VI
THE AVERAGE PREDICTION ERROR FOR DIFFERENT MUSCLE COMBINATIONS AT A PREDICTION TIME OF 40 ms

Group	Number of features	RMSE (°)	adjusted R^2
VLST	10	4.55	0.91
TAMG	4	5.41	0.87
VLTA	7	6.95	0.80
STMG	7	5.78	0.85
VLMG	7	6.67	0.81
STTA	7	4.55	0.90
ALL	14	3.55	0.95

IV. Discussion

Continuous prediction of lower limb joint angles is crucial for the application of lower limb exoskeletons. Gui et al. [48] noted that control schemes based on neuromuscular principles can significantly enhance the control efficacy of prosthetics. This study introduces an LSTM model integrated with an attention mechanism for the continuous prediction of knee joint angles. Additionally, a feature selection algorithm based on XGBoost is proposed to identify efficient features, eliminate redundancy, and reduce the dimensionality of input data. Experimental results on the test set indicate that the average RMSE of the model in predicting the knee joint angle in the future 40 ms is 3.55°. Visualization of the results demonstrates that the proposed algorithm has the ability of continuous smooth and low error prediction, which shows the potential to realize the continuous smooth control of the exoskeleton robot with lower limbs.

For successful knee angle prediction based on sEMG signals, the selection of extracted sEMG features should be carefully considered. However, existing studies utilize feature sets that contain a large number of redundant features [19], [49]. Here, we employed the XGBoost algorithm to assess the contribution of seven common sEMG features to joint angle predictions and to determine the optimal feature set. The findings reveal that the feature combination of WL and MDF

TABLE VII
PREVIOUS WORK ON THE JOINT ANGLE PREDICTION TASK

Author	Number of sensors	Sensors	Methods	MAE (°)	RMSE (°)	adjusted R^2	CC
Li *et al.* [50]	12	sEMG sensors, Vicon 3D angle system	MLP, Savitzky-Golay filter	–	7.72	–	–
Sun *et al.* [28]	16	IMU sensors, sEMG sensors	CNN-BiLSTM	–	4.07	0.95	0.98
Wang *et al.* [18]	4	sEMG sensors	dSTA-EMGNet	2.60	4.23	–	–
Liang *et al.* [51]	5	sEMG sensors	TCCT	–	3.79	0.96	0.98
Wang *et al.* [52]	5	sEMG sensors	Wavelet transform and ELMAN network	–	4.16	–	–
Deng *et al.* [53]	6	sEMG sensors	PCA and ELM	–	8.47	–	0.95
Zhao *et al.* [54]	4	terrain images IMU sensors	SFTIK	–	3.45	–	0.97
This work	4	sEMG sensors	CAELSTM	2.47	3.55	0.95	0.98

appears most frequently, indicating a substantial correlation with knee joint angles during gait. The VL and ST muscles exhibit more influential sEMG features, highlighting their primary roles during walking. The VL muscle is pivotal for knee extension and hip flexion, maintaining an upright posture, while the ST aids in propelling the body forward during walking. In addition, different muscle combinations also have an important impact on joint angle prediction, for which we designed channel drop experiments to evaluate six different combinations of four muscles from the anterior and posterior sides of the thigh or shank. Table VI shows that the CAELSTM model has the best performance for knee angle prediction when all four muscles are used, while different muscle combinations have different effects on the prediction performance. Therefore, the prediction accuracy of joint angles is not only related to the number of sEMG channels but also to the combination of different muscles of the lower limb. This provides a reference for the practical application of lower limb exoskeleton robots to select the appropriate electrode channel configuration for patients with different conditions.

During walking, the amplitude of sEMG of human lower limbs changes with time, and it is an effective method to use past data to predict the knee joint angle. LSTM is widely used because of its elaborate structure designed for time series, which is suitable for predicting human motion variables. In the feature extraction stage, a hybrid structure of CNN and LSTM is used to extract hidden features at different time steps. In the feature refinement stage, FGCAM was used to perform global and local fine-grained interaction, and FDECAM introduced frequency domain information to enhance the prediction ability of the model. In the prediction output stage, the dependency relationship between the current time step vector and the historical information is learned by the HFFM module to obtain a more robust feature vector. Ablation experiments verify the effectiveness of the different structures of the model. In addition, the prediction accuracy of the proposed method not only exceeds that of traditional ML methods, but also exceeds that of DL models, including CNN, LSTM, TCN, and CNN-BilSTM.

In addition, we further verify the real-time performance of the proposed method. Experiments demonstrated that the model achieves an average inference latency of 33.16 ms on our testing platform, representing a 47% reduction compared to the 62.47 ms latency reported in [28]. Leveraging the physiological property that sEMG signals precede actual movement by 30∼150 ms, our model predicts knee joint angles 40 ms in advance, reserving a time window for subsequent control commands. Combining the algorithmic delay with the prediction lead time, the total response time remains well below the typical mechanical system delay threshold, thereby fully satisfying real-time requirements and leaving sufficient margin for other mechanical delays.

To further evaluate the performance of the proposed method, we compare and analyze the results of other researchers in recent years on the task of sEMG-based joint angle prediction, as shown in Table VII. Multisensor fusion methods are usually more accurate than single-sensor signal prediction. For example, Sun et al. [28] achieved better prediction performance than single sensor signals by fusing the features of sEMG and IMU signals. Using only sEMG signal input, our CAELSTM model achieves comparable prediction accuracy to the multisensor fusion method. Compared to the transformer-based prediction model used by Liang et al. [51], which is more complex, our proposed method demonstrates superior efficiency and achieves higher prediction accuracy.

A key challenge in sEMG-based human movement intention prediction is the generalization across subjects. Transfer learning can be a solution by extracting general domain knowledge from other subjects' data and applying it to the target subject. Our model's evaluation has been done offline. But in real exoskeleton use, more noise and abnormal sEMG signals may occur. Despite our method's good offline performance, its online effectiveness needs further assessment and adjustment when handling abnormal signals.

V. CONCLUSION

In this study, we aimed to accurately predict knee joint angles using only sEMG signals to generate joint angle trajectories for LLERos, thereby enhancing human–machine interaction. We proposed a multistage DL network, CAELSTM, which incorporates an attention mechanism to capture the correlation features of historical and state information in sEMG signals. A feature selection algorithm based on XGBoost was introduced to eliminate redundant sEMG features and reduce input dimensionality. Experimental results indicated no significant difference in predictive performance before and after feature removal. On our self-collected dataset, CAELSTM outperformed baseline models with lower prediction errors. We also investigated the impact of various prediction intervals, ranging from 40 to 200 ms, on prediction accuracy. The results demonstrated that as the lead time for prediction increased, the model's predictive performance deteriorated, with higher RMSE observed for lead times more than 200 ms. This reveals the boundary for compensating mechanical transmission delays. These findings underscore the predictive performance and adaptability of our proposed framework, highlighting its potential application in LLERo control to improve the efficiency of patient rehabilitation training.

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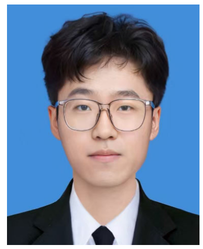
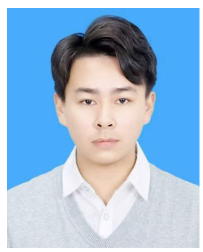
Liuyi Ling received the Ph.D. degree in optics from the University of Chinese Academy of Sciences, Hefei, China, in 2013.

He is currently a Professor with Anhui University of Science and Technology, Huainan, China. His current research interests include detection technology and information processing technology.



Liyu Wei received the B.S. degree in Internet of Things engineering from Nanjing University of Information Science and Technology, Nanjing, China, in 2022. He is currently pursuing the master's degree in artificial intelligence from Anhui University of Science and Technology, Huainan, China.

His research focuses on the intersection of wearable sensors and artificial intelligence. His interests lie in deep learning, wearable sensors, and embedded systems.



Bin Feng Bin Feng received the B.S. degree in electrical engineering and intelligent control from the School of Electrical and Information Engineering, Anhui University of Science and Technology, Huainan, China, in 2021. He is currently pursuing the Ph.D. degree in artificial intelligence with Anhui University of Science and Technology.

His main research direction is robot motion control, motion planning, and perception.

Zhipeng Yu received the Ph.D. degree from the University of Science and Technology of China, Hefei, China, in 2023.

Currently, he is a Teaching Assistant at the Department of Robot Engineering, School of Artificial Intelligence, Anhui University of Science and Technology, Huainan, Anhui, China. He is mainly engaged in the teaching and research work of rehabilitation robot, human–computer interaction, robot trajectory planning, and intelligent control.

Long Wang received the B.S. degree from Anhui University of Science and Technology, Huainan, China, in 2022, where he is currently pursuing the master's degree in artificial intelligence.

His research interests include machine learning, signal processing, and time series forecasting.